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 Studierichting: Informatica
 Oefeningenreeks 6A nr. 25.4

$(0, 0, 1, 5, 15, 35, 70, \dots)$

	0	0	1	5	15	35	70
	0	1	4	10	20	35	
		1	3	6	10	15	
			2	3	4	5	
			1	1	1		

$\Rightarrow 4^{\text{de}} \text{ orde}$

	0	0	1	5	15	35	70	126
	0	1	4	10	20	35	56	
		1	3	6	10	15	21	
			2	3	4	5	6	
			1	1	1	1		

$\Rightarrow x_8 = 126$

$\Rightarrow x_n = ???$

$$\Rightarrow \begin{cases} a + b + c + d + e = 0 \\ 16a + 8b + 4c + 2d + e = 0 \\ 81a + 27b + 9c + 3d + e = 1 \\ 256a + 64b + 16c + 4d + e = 5 \\ 625a + 125b + 25c + 5d + e = 15 \end{cases}$$

$$\Rightarrow \begin{cases} 16a + 7b + 3c + d = 0 \\ 65a + 18b + 5c + d = 1 \\ 175a + 37b + 7c + d = 4 \\ 369a + 61b + 9c + d = 10 \end{cases}$$

$$\Rightarrow \begin{cases} 8a + 12b + 2c = 1 \\ 110a + 18b + 2c = 3 \\ 194a + 24b + 2c = 6 \end{cases}$$

$$\Rightarrow \begin{cases} 60a + 6b = 2 \\ 84a + 6b = 3 \end{cases}$$

$\Rightarrow 24a = 1$

$$a = \frac{1}{24}, b = -\frac{1}{12}, c = -\frac{1}{24}, d = \frac{1}{12}, e = 0$$

$$x_n = \frac{1}{24}n^4 - \frac{1}{12}n^3 - \frac{1}{24}n^2 + \frac{1}{12}n$$

$$= \frac{n^4 - 2n^3 - n^2 + 2n}{24}$$

$$= \frac{n(n^3 - 2n^2 - n + 2)}{24}$$

$$\begin{aligned}
&= \frac{n(n^2(n-2) - 1(n-2))}{24} \\
&= \frac{n((n^2-1)(n-2))}{24} \\
&= \frac{n(n-1)(n+1)(n-2)}{24}
\end{aligned}$$

References